Electrical Power and Energy Systems 171 (2025) 110978

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

An enhanced modal controllability-observability driven power control scheme for active suppression of rotor in-plane loading in large-scale PMSG-based wind turbine systems

Ali Safaeinejad, Mohsen Rahimi *

Department of Electrical and Computer Engineering, University of Kashan, Kashan, Iran

ARTICLE INFO

Keywords:
Rotor in-plane motions
Active mitigation control
Auxiliary damping terms
Modification site

ABSTRACT

Many reliability challenges in modern large-scale wind turbines (WTs) arise from rotor in-plane loads along the rotor plane direction, commonly referred to as rotor in-plane modes. These detrimental loads are caused by the weakly damped flexibilities of the blade edgewise, drivetrain torsion, and tower side-side dynamics, resulting in noticeable operation and maintenance costs. This paper delves into the design of an active mitigation control for rotor in-plane mechanical-structural loads in Permanent Magnet Synchronous Generator (PMSG) WTs. This control scheme is designed via a high-accuracy model accounting for the dynamic coupling between the electrical, mechanical, structural, and aerodynamic parts. To enhance the intrinsic damping of these loads, auxiliary terms are incorporated into the control system. The precise determination of appropriate feedback signals is achieved through the modal observability analysis, while the identification of the suitable points for applying them is elaborated by the controllability analysis. The proposed control strategy is implemented within the power control assigned to the grid side converter, where the reference power is updated by adding three auxiliary terms associated with the damping improvement of the blade edgewise, drivetrain, and tower side-side deflections. Finally, simulation results in both the frequency and time scopes approve the deservedness of the proposed scheme in restraining the rotor in-plane loads.

1. Introduction

The cutting-edge technologies and evolving trends in modern wind turbines (WTs) are oriented towards multi-megawatt scales, along with large structural components. This meets the growing demand for productivity and efficiency expectations from wind farms. The large-scale WTs are commercially manufactured with lightweight and high-strength materials, leading to low stiffness and light damping in their mechanical and structural modes [1]. Under operational conditions, these modes are constantly exposed to aerodynamic and electrical stimuli, causing adverse vibrations and loads in the costly and huge structures of the WTs such as blades, tower and drivetrain components, and these can conceive a menace to their service life [2]. In general, fatigue loads in WTs are broadly divided into two directions: rotor in-plane and rotor out-of-plane. Out of the two, rotor in-plane loads, including drivetrain torsional, blade edgewise, and tower side-side deflection lightly damped modes, cause remarkable downtimes of the huge WT structures [3].

To address these challenges, accurate yet computationally efficient modeling of turbine dynamics is essential for control design. In the current study, the blade and tower dynamics are modeled using simplified yet physically meaningful representations, with the blade treated as a cantilever beam undergoing edgewise bending and the tower modeled as a lumped-mass system exhibiting side-side flexibility. The model retains only the dominant first-order structural modes, which are most critical in the fatigue behavior of large-scale WTs. Aerodynamic forces are computed using a corrected Blade Element Momentum (BEM) approach, accounting for both axial and tangential induction effects. Compared to high-fidelity computational fluid dynamics (CFD)-based models [4,5] which solve the Navier-Stokes equations to resolve flow-structure interactions in full detail, the proposed analytical modeling approach offers significantly lower computational cost while retaining the essential physical characteristics needed for controller synthesis and stability analysis. On the other hand, while data-driven models based on artificial intelligence can provide fast approximations [6,7], their accuracy heavily depends on the quality of training data and they often

* Corresponding author.
E-mail addresses: a.safaeinejad@grad.kashanu.ac.ir (A. Safaeinejad), mrahimi@kashanu.ac.ir (M. Rahimi).

https://doi.org/10.1016/j.ijepes.2025.110978
Received 17 March 2025; Received in revised form 15 July 2025; Accepted 3 August 2025
Available online 13 August 2025

lack interpretability. Our physics-based framework combines small-signal analysis with Lyapunov stability verification, offering a theoretically grounded solution for active load mitigation.

This paper first addresses the dynamic behavior of the rotor in-plane modes in Permanent Magnet Synchronous Generator (PMSG) WTs, and then, proposes an active mitigation control approach to alleviate the fatigue loads imposed on the WT drivetrain, blades and tower. In order to lessen the operational and maintenance costs and enhance the reliability in large-scale WTs, the rotor in-plane loads should be efficiently mitigated. Among the control strategies for fatigue load mitigation, active approaches have been more interested than classic passive methods. In active approaches, an opposing force is adapted to suppress variable loads by feeding vibration signals to the WT control system. These forces, indeed, counteract detrimental vibrations in the blades, tower, and drivetrain elements [1–8]. To the best knowledge of the authors, no analytical research is available in the literature addressing the active mitigation of all types of rotor in-plane loads considering the electro-mechanical-structural interactions and optimum design of the active control strategy.

Active mitigation approaches for torsional oscillations are categorized as traditional and modern techniques. In traditional techniques, a damping torque is built using the torsional harmonic content, extracted from the generator speed feedback via a tuned filter [9,10]. Modern techniques are based on the premise that the torsional oscillations originate from the difference in rotational speeds between the high-speed and low-speed shafts, offering superior performance over traditional techniques under the existence of uncertainty conditions in the mechanical system. In [10–14], the damping action is achieved by injecting the difference between the rotational speeds of the turbine and generator into the original generator torque. Nevertheless, in these works, any analysis about the selection of the most effective injecting site and optimal feedback signals for providing the damping torque has not been done.

In the context of blade edgewise vibrations, Ref. [15] shows that the grid voltage dips can excite the edgewise deflections, however it does not present any suggestion to alleviate them, furthermore, the influence of wind variations on them has not been investigated. In [16]- [17], an active mitigating control is designed based on a mathematical representation of aerodynamic loads and a reduced-order dynamic model for the edgewise motions. These control schemes rely on a pair of active tendon actuators which should be embedded inside each blade, imposing extra cost and reliability issues.

In the literature, less analytical work has been performed on side-side fatigue mitigation. In [18], for an offshore WT, it has been assumed that the WT operates in speed controlling mode and the tower first mode frequency interferes with the WT operational speed region. Based on this assumption, a speed exclusion zone algorithm is proposed to avoid the side-side resonance. However, the effect of power-curve manipulation on energy efficiency and other mechanical-structural responses has not been reported. In [3], for a PMSG-based WT in the speed control mode, active damping of side-side vibrations is carried out via modifying the inner control loop of the machine side converter. The damping torque of this method is calculated based on a sophisticated structural model, and thus, its effectiveness may be affected when subjected to parametric uncertainty. Ref. [19] suggests the use of an adaptive internal model-based controller to diminish the side-side motions via the pitch control system. This method has practical limitations such as identifying the side-side natural frequency and measuring the blades moments. Moreover, the analytical basis of how the pitch controller dynamically impacts the side-side vibrations have not been represented.

Motivated by the literature context and research gaps outlined above, the ongoing research proposes an active mitigation control scheme to effectively suppress the rotor in-plane loads, including WT shaft torsional, blade edgewise, and tower side-side vibrations, without using any extra actuator in the WT structures. In this research, the WT is based on a PMSG operated in the power control mode, consistent with practical applications. The control design procedure proceeds based on an in-depth insight into the dynamic behavior of the bending and shafting modes. The key novelties and contributions of this research are summarized as follows: (1) Establishing a high-fidelity model to investigate the dynamic coupling between electrical, mechanical, aerodynamic, and structural sections in the WT (2) Finding the intrinsic oscillatory modes in the mechanical-structural sections of the study WT and identifying the predominant factors affecting these modes based on small-signal analysis (3) Examining different modification sites and feedback signals to identify the best injecting point and effective control feedbacks for creating auxiliary damping power using modal controllability and observability analyses (4) Proposing an active mitigation control approach for suppressing the rotor in-plane loads by integrating the appropriate auxiliary terms into the grid-side converter (GSC) control and matching the pitch angle controller with the proposed control scheme.

2. Weight mathematical modelling

This Section aims to present the mathematical modelling for the PMSG-based WT, by which, the rotor in-plane loads can be investigated. This model consists of the mechanical, aerodynamic, structural, control and generator subsystems.

2.1. Aerodynamic modelling

Considering the impact of the blade passing phenomenon (BPPh) on the WT energy conversion, the mechanical torque harvested from the wind flow for a horizontal-axis three-bladed, upwind-turbine can be expressed as [20]:

$$T_t = T_m + \widetilde{T}_{BP} \tag{1}$$

$$T_m = \frac{1}{2}\rho\pi R^2 (V_w - \dot{x}_{nd})^3 C_p(\lambda, \beta)/\omega_b \tag{2}$$

$$\widetilde{T}_{BP} = \frac{2}{mV_H}\left[v_{eq_{ws}} + v_{eq_{ts}} + (1-m)V_H\right]T_m \tag{3}$$

where T_m and $\widetilde{T}_{BP}$ are the mean value of the turbine torque and its oscillatory term caused by BPPh, respectively. Further, V_w, R, ρ, ω_b, and $\dot{x}_{nd}$ represent the spatial average wind speed, length of blades, air density, rotational speed of the blades, and tower top velocity in the for-aft direction, respectively. Also, C_p as the power coefficient of the blades, relies on the pitch angle of the blades, β, and tip speed ratio, λ, where λ, as the tip speed ratio, is obtained as

$$\lambda = \frac{R\omega_b}{(V_w - \dot{x}_{nd})} \tag{4}$$

In addition, m in (3) denotes the proportional ratio between spatial wind speed, V_w, and wind speed at the hub height, V_H, i.e. $m = V_w/V_H$. Also, $v_{eq_{ts}}$ and $v_{eq_{ws}}$, indicating the equivalent wind speeds corresponding to the tower shadow and wind shear effects, are expressed as

$$v_{eq_{ts}} = \frac{mV_H}{3R^2}\sum_{b=1}^{3}\left[\frac{a^2}{\sin^2\theta_b}\ln\left(\frac{R^2\sin^2\theta_b}{x^2}+1\right) - \frac{2a^2R^2}{R^2\sin^2\theta_b + x^2}\right] \tag{5}$$

$$v_{eq_{ws}} = V_H\left[\frac{\alpha(\alpha-1)}{8}\left(\frac{R}{H}\right)^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{60}\left(\frac{R}{H}\right)^3\cos 3\theta_b\right] \tag{6}$$

where θ_b, x, a, H, and α are the blade azimuthal angle, blade origin from the tower midline, tower radius, height of the hub, and empirical wind shear exponent, respectively.

2.2. Mechanical-Structural assembly

Due to the complexity and multi-interaction nature of the WT

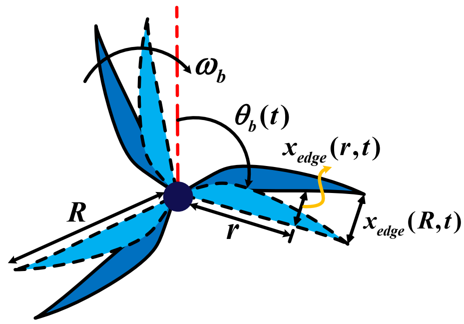

Fig. 1. Visual representation of the blade edgewise bending.

vibrations, to elaborate the dynamic features of the rotor in-plane motions, the intrinsic modes as well as corresponding forces (or torques) acting on them should be taken into account. In the following, dynamic modeling of the blade bending, drivetrain flexibilities, and tower vibrations from the perspective of the rotor in-plane motions is presented.

2.2.1. Modelling of the blade edgewise motions

Fig. 1 illustrates the visual representation of the blade edgewise motions, where the edgewise motions, as function of the time and length of the blades, are displayed by $x_{edge}(r,t)$ symbol [16]. In order to establish a dynamic model for disclosing the edgewise bending modes of the blades, the blade construction is normally considered as flexible cantilever beams [17]. Since edgewise resonant loads are primarily influenced by the first mode of edgewise vibrations, mathematical analyses generally focus only on the fundamental edgewise frequency, assuming that each beam oscillates at its first-mode frequency [15,17]. Therefore, by utilizing the potential and kinetic energies of the blades and ignoring interactions between the blades and tower, the swing equation for the blade edgewise vibrations can be considered as a mass-spring-damper system given by [15]

$$M_{edge}\ddot{x}_{edge}(t) + D_{edge}\dot{x}_{edge}(t) + K_{edge}x_{edge}(t) = F_{\mathrm{tang}}(t) \tag{7}$$

$$F_{\mathrm{tang}}(t) = F_{wind}(t) + F_{grav}(t) + F_{inert}(t) \tag{8}$$

where, M_{edge}, K_{edge}, and D_{edge} are the edgewise mass, stiffness and coefficients, respectively, based on the structural specifications of the blades. Also, $x_{edge}(t)$ represents the edgewise bending motion, and $F_{tang}(t)$ implies on the tangential force applied to the blade airfoils, consisting of three components: the aerodynamic tangential force due to wind loading, $F_{wind}(t)$, gravitational force projected onto the edgewise direction of the blades, $F_{grav}(t)$, and inertial force associated with the edgewise acceleration of the blade, $F_{inert}(t)$. Based on the (7), and the specifications of the WT system under study, given in the Appendix, the 1st order edgewise resonant frequency is obtained to be 1.95 Hz.

Among the driving input forces in the blade in-plane direction, the tangential aerodynamic force due to wind loading plays the dominant role in stimulating the edgewise vibrations. To develop a high-fidelity dynamic model for the wind-induced force, a collective pitch angle and homogeneous rotor body are assumed, implying that all three blades undergo identical edgewise deflections. The aerodynamic behavior of the blade airfoils depends on the incoming wind speed, blade rotational speed, and geometrical attributes of the blades, and is specified by the corrected form of the (BEM) theory [13,21]. As illustrated in Fig. 2, and under the assumption of negligible aerodynamic interaction between airfoils, the tangential force applied on each airfoil with chord thickness $d(r)$ and chord length $c(r)$ at the radial distance r from the hub, is obtained as

$$F_{wind}(r,t) = F_L(r,t)\sin(\phi) - F_D(r,t)\cos(\phi) \tag{9}$$

Where Φ, F_L, and F_D are the inflow angle, lift force, and drag force, respectively.

Although the gravitational and inertial components are less significant than the aerodynamic excitation, they are included to ensure an accurate representation of the edgewise dynamic response of the blades. The gravitational component varies periodically with the azimuthal position of the blade, leading to low-frequency oscillations in the edgewise direction. Furthermore, the inertial force due to blade acceleration along the edgewise direction also contributes to the overall tangential excitation. The mathematic demonstration of these equations can be described as follows

$$F_g(t) = m_r g\cos(\theta_b) \tag{10}$$

$$F_a(t) = -m_r a_b(t) \tag{11}$$

where m_r denotes the mass of the blade section at radial position r, g is the gravitational acceleration, and a_b indicates the acceleration of the blade section, $\dot{\omega}_b$.

Considering the uniform wind turbulence across the rotor area, the linearized representation of the tangential force near an operating point pertaining to the blade-hub shaft twist angle, $\theta_{sh\text{-}bh0}$, blade rotational speed, ω_{b0}, pitch angle, β_0, wind speed, V_{w0}, and azimuthal angle, θ_{b0}, can be derived as

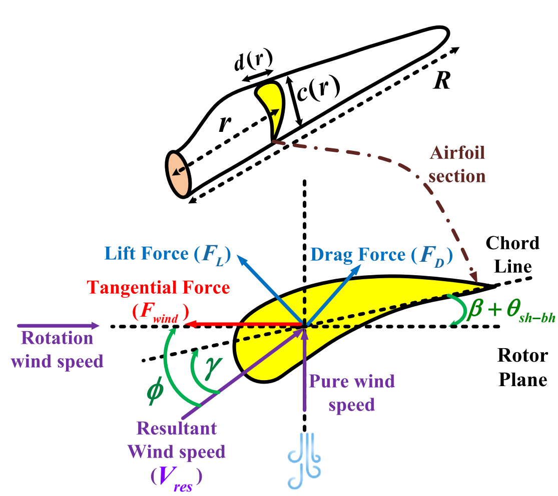

Fig. 2. Demonstration of the aerodynamic forces and velocities, and flow angles on a typical airfoil profile.

$$\Delta f_{\tan} = \left[(f_{L_0}\cos\phi_0 + f_{D_0}\sin\phi_0 + \frac{1}{2}\rho A V_{res_0}^2(\sin\phi_0\frac{\partial C_L}{\partial\gamma} - \cos\phi_0\frac{\partial C_D}{\partial\gamma}))\frac{\partial\phi}{\partial\omega_b} + (\frac{1}{2}\rho A 2V_{res_0}(\mathrm{C}_{L_0}\sin\phi_0 - \mathrm{C}_{D_0}\cos\phi_0))\frac{\partial V_{res}}{\partial\omega_b}\right]\Delta\omega_b + \left[(\frac{1}{2}\rho A V_{res_0}^2(\cos\phi_0\frac{\partial C_D}{\partial\gamma} - \sin\phi_0\frac{\partial C_L}{\partial\gamma})\right](\Delta\beta + \Delta\theta_{sh-bh}) + \left[(f_{L_0}\cos\phi_0 + f_{D_0}\sin\phi_0 + \frac{1}{2}\rho A V_{res_0}^2(\sin\phi_0\frac{\partial C_L}{\partial\gamma} - \cos\phi_0\frac{\partial C_D}{\partial\gamma}))\frac{\partial\phi}{\partial V_w} + \frac{1}{2}\rho A 2V_{res_0}(C_{L_0}\sin\phi_0 - C_{D_0}\cos\phi_0))\frac{\partial V_{res}}{\partial V_w}\right]\Delta V_w - \left[m_r g\sin\theta_{b_0}\right]\Delta\theta_b \tag{12}$$

where V_{res_0}, C_{D_0}, C_{L_0}, ϕ_0, f_{D_0}, f_{L_0}, and θ_{b_0} refer to the resultant speed, drag coefficient, lift coefficient, inflow angle, drag force, lift force, and azimuthal angle respectively, at a specified operating point. It should be noted that the inertial force component is not included in the linearized representation of the tangential force due to its limited influence on the edgewise excitation, as well as to avoid unnecessary complexity in the formulation. Nevertheless, it is fully considered in the subsequent nonlinear simulation studies.

2.2.2. Modeling of the tower side-side vibrations

To describe the tower side-side vibrations, the tower structure is first considered as a mass-free cantilever beam, with the rotor and nacelle attached at its free end on the top of the tower. Subsequently, the tower is modeled as a lumped-mass system, with its total mass concentrated at the top. The effective mass is calculated as the sum of the masses of the tower, rotor, and nacelle. Fig. 3 portrays visual illustration of the side-side vibrations, in which the side-side deflections are represented by the time-dependent variable $x_{side}(t)$. Similar to the blade edgewise vibrations, the 1st order mode of the tower side-side vibrations plays the crucial role in side-side resonant loading, in which, the swing equation of the tower in the rotor in-plane frame can be written as [18,22], and [23].

$$M_{tow}\ddot{x}_{side}(t) + D_{tow}\dot{x}_{side}(t) + K_{tow}x_{side}(t) = f_{nac} \tag{13}$$

where K_{tow}, D_{tow} and M_{tow} are the stiffness coefficient, damping coefficient, and effective mass of the tower, respectively, also H_{tow} denotes the height of the tower. Besides, f_{nac} denotes the nacelle force and represents the counterforce imposed by the generator on the nacelle and tower, defined as [24]

$$f_{nac} = \frac{3n_g T_e}{2H_{tow}} \tag{14}$$

where T_e and n_g imply on the electromagnetic torque and gear ratio and, respectively. According to (13) and the studied WT characteristics given in the Appendix, the tower side-side eigenfrequency is calculated as 2π × 0.32 rad/sec.

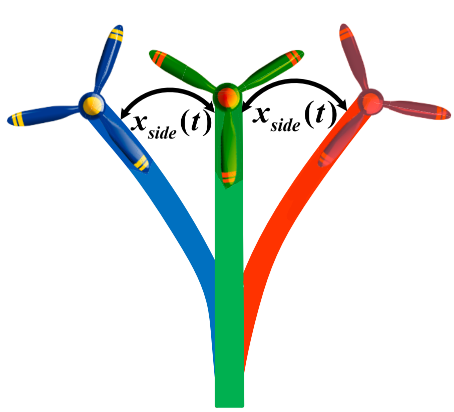

Fig. 3. Visual demonstration of the tower side-side deflections.

2.2.3. Modelling of the drivetrain system

The drivetrain assembly comprises the low speed and high-speed shafts, hub, gearbox, blades, and generator. Referring Section 2.2.1, the edgewise bending are linked to the drive-train dynamics via the blade-hub shaft twist angle, $\theta_{sh\text{-}bh}$, and also the blade rotational speed, ω_b, and this means that the blades flexibilities can affect the drivetrain dynamics. Thus, to simultaneously consider the drive-train and blades flexibilities, the dynamic coupling between the drivetrain and the blades must be taken into consideration. Attributing the blade edgewise bending mode to the tip blade airfoils, the rotor dynamics can be split into two segments, the blades and the hub [16,17,25]. Hence, the drivetrain structure can be described as three-mass model whose demonstration referred to the high-speed shaft is demonstrated in Fig. 4. Based on this arrangement, the drivetrain includes three segregate masses coupled together via the intermediate inertialess shafts with the related stiffness and damping.

For simplicity, the linearized representation of the dynamic equations governing the three-mass drivetrain model are expressed in per-unit as follows:

$$\Delta T_t - \Delta T_{sh-bh} - D_b\Delta\omega_b = 2H_b\frac{d\Delta\omega_b}{dt} \tag{15}$$

$$\Delta T_{sh-bh} - \Delta T_{sh-hg} - D_h\Delta\omega_h = 2H_h\frac{d\Delta\omega_h}{dt} \tag{16}$$

$$\Delta T_{sh-hg} + \Delta T_e - D_g\Delta\omega_g = 2H_g\frac{d\Delta\omega_g}{dt} \tag{17}$$

$$\Delta T_{sh-bh} = k_{bh}\Delta\theta_{sh-bh} + D_{bh}(\Delta\omega_b - \Delta\omega_h) \tag{18}$$

$$\Delta T_{sh-hg} = k_{hg}\Delta\theta_{sh-hg} + D_{hg}(\Delta\omega_h - \Delta\omega_g) \tag{19}$$

$$\frac{d\Delta\theta_{sh-bh}}{dt} = \omega_B(\Delta\omega_b - \Delta\omega_h) \tag{20}$$

$$\frac{d\Delta\theta_{sh-hg}}{dt} = \omega_B(\Delta\omega_h - \Delta\omega_g) \tag{21}$$

where Δ illustrate the small variations around a specified operating point, $\theta_{sh\text{-}bh}$ and $\theta_{sh\text{-}hg}$ imply on the shaft twist angles at blade-hub and

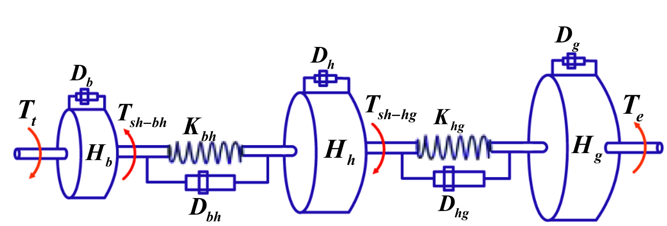

Fig. 4. The three-mass drivetrain model demonstration.

hub-generator interfaces, respectively, ω_h and ω_g indicate the rotational speeds of the hub and generator, respectively, also T_e and T_t indicate the torques of the generator and turbine, respectively. $T_{sh\text{-}hg}$ and $T_{sh\text{-}bh}$ are the shaft torsional torques corresponding to the hub-generator and blade-hub interfaces, respectively. The inertia constants associated with the generator, blades, and hub are signified by H_g, H_b, and H_h, respectively. Also, the shaft stiffness coefficients related to the blade-hub and hub-generator interfaces are denoted by k_{bh} and k_{hg}, respectively. Also, the mutual damping coefficients of the hub-generator and blade-hub shafts are symbolized by D_{hg} and D_{bh}, respectively. Further, the self-damping coefficients related to the hub, generator, and blades are indicated by D_h, D_g, and D_b, respectively. Also, ω_B represents the base value of the electrical angular frequency in (rad/sec). From the studied WT parameters given in the Appendix, the 1st and 2nd orders of the drivetrain natural frequencies are calculated as 2.29 Hz and 13.34 Hz, respectively. It is perceived that the 1st order torsional resonant frequency, i.e. 2.29 Hz, is very near to the edgewise natural frequency, which was 1.95 Hz.

2.3. PMSG dynamic model

The dynamic equations for a non-salient pole PMSG in the synchronous d-q reference frame aligned with the rotating rotor are given by (22)-(23) [26].

$$v_{sdq} = R_s i_{sdq} + j\omega_g \psi_{sdq} + \frac{1}{\omega_B}\frac{d\psi_{sdq}}{dt} \tag{22}$$

$$\psi_{sdq} = L_s i_{sqd} + \psi_{pm} \tag{23}$$

In (22)-(23), all parameters and variables are described in per-unit, with the positive flow of the stator current considered toward the generator windings. ψ, v, and i denote the flux, voltage, and current, and subscript s implies on the stator quantities, R_s and L_s indicate the stator resistance and inductance, ψ_{pm} refers to the magnitude of the stator flux linkage resulting from the rotor permanent magnet. Under steady state conditions and ignoring the generator losses, the active power produced by the stator windings and electromagnetic torque can be expressed as [26]:

$$P_s = -\psi_{pm}\omega_g i_{sq} \tag{24}$$

$$T_e = \psi_{pm} i_{sq} \tag{25}$$

2.4. Basic control system for the WT

This section addresses the control structure of the power electronic converters, consisting of the grid-side converter (GSC), machine-side converter (MSC), and pitch angle control system.

2.4.1. Control system of the GSC and MSC

Fig. 5 shows the overall block diagram of the control arrangement for the power electronic converters in the PMSG-based WT system under study. In this configuration, the MSC is responsible for regulating the DC-link voltage, while the GSC controls the active power delivered to the grid. As demonstrated in [21], this control structure provides desirable low-voltage ride-through LVRT capability, even under severe three-phase voltage dips. The active power reference, $P_{g,ref}$, is determined by a predefined power-speed curve derived from aerodynamic characteristics of the WT blades. This curve generally comprises two regions:

Optimum region: from the cut-in wind speed up to the rated wind speed, where the turbine maximizes the energy capture at a fixed blade pitch angle of 0°.

The full-load region: from the rated wind speed to the cut-out wind speed, where the power output is regulated at its nominal value through the pitch actuation.

The reference values for the inner current control loops, $i_{sq,ref}$ and $i_{gd,ref}$, are determined by the outer DC-link voltage and the active power controllers, respectively. In the GSC, v_{cd}, v_{cq}, i_{gd} and i_{cq} denote the dq-axis components of the grid voltage and current, defined in the synchronous reference frame aligned with the grid voltage. The phase and frequency of the grid voltage are detected by the phase locked loop (PLL).

The controller gains in both the inner and outer loops are tuned using the pole-zero cancellation technique or a second-order system design approach with appropriate bandwidth and damping ratio, as expressed in our previous work [26]. This ensures satisfactory dynamic response and closed-loop stability across a wide operating range.

2.4.2. Pitch angle control system

The primary task of the pitch angle control system is to preserve the WT structure against excessive mechanical loads when the wind field goes beyond its upper limit [27]. Fig. 6 shows the closed-loop arrangement of the pitch control system adopted from [28], where τ_β, β_{ref}, $\omega_{g,ref}$, and β are the time constant of the pitch actuator, pitch angle command, allowed operating speed of the turbine, and actual pitch, respectively. The procedure for designing the pitch controller is described in detail later.

3. Small signal stability analysis for the studied WT system with the basic control arrangement

The dynamic representation of the whole study WT system, stated in

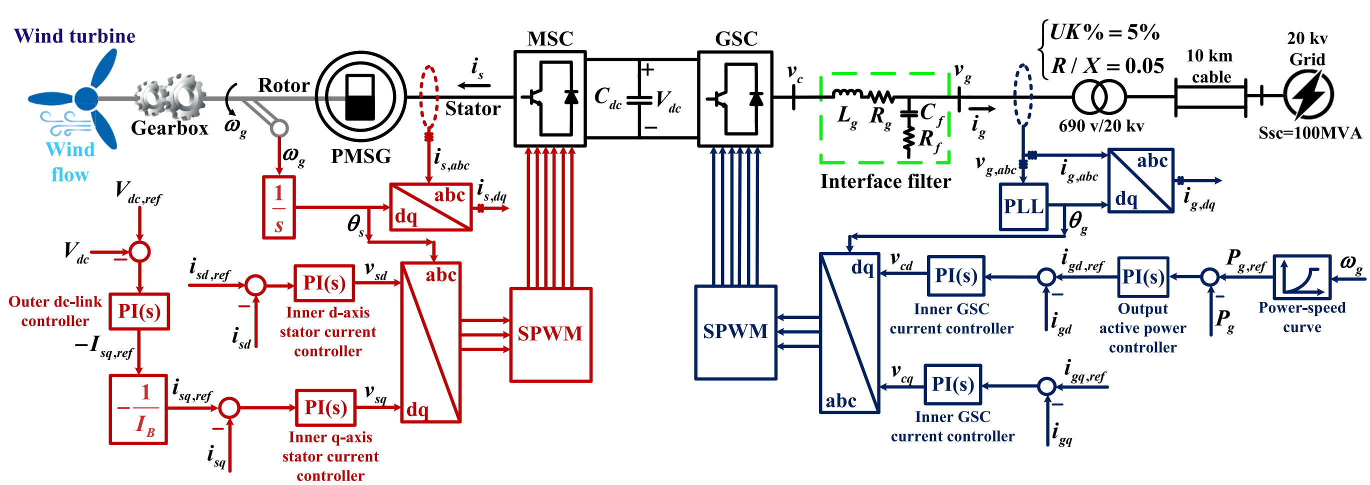

Fig. 5. The block diagram overview of the MSC and GSC for the PMSG-based WT in the basic control pattern.

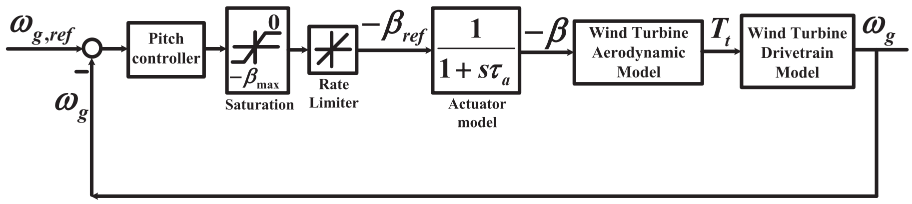

Fig. 6. Closed-loop arrangement of the pitch angle system.

Table 1
Eigenvalues and Dominant State Variables For the studied WT With the Basic Control System.

Modes	Frequency (Hz)	Damping ratio	Dominant state variables
λ_1 = - 677.44	–	1	i_{sd}
λ_2 = - 18.13	–	1	$x_{i_{sd}}$
$\lambda_3 = -586.93$	–	1	i_{sq}
$\lambda_{4,5} = -628.32 \pm 0j$	–	1	i_{gd}, i_{gq}
$\lambda_{6,7}$ = - 45.92 ± 32.46j	5.17	0.82	$V_{dc}, x_{V_{dc}}$
$\lambda_8 = -18.18$	–	1	$x_{i_{sq}}$
$\lambda_9 = -62.83$	–	1	x_{P_g}
$\lambda_{10,11} = -6.28 \pm 0j$	–	1	$x_{i_{gd}}, x_{i_{gq}}$
$\lambda_{12} = -9.05$	–	1	β
$\lambda_{13,14} = -0.76 \pm 0.46j$	0.068	0.88	$x_\beta, \omega_b, \theta_b$
$\lambda_{15,16} = -0.27 \pm 14.72j$	2.34	0.0183	θ_{sh-bh}, ω_g
$\lambda_{17,18}$ = - 5.34 ± 83.74j	13.35	0.0636	θ_{sh-hg}, ω_h
$\lambda_{19,20} = -0.06 \pm 12.25j$	1.95	0.005	$x_{edge}, \dot{x}_{edge}$
$\lambda_{21,22} = -0.02 \pm 2.02j$	0.32	0.01	$x_{side}, \dot{x}_{side}$

Section 2, can be described as

$$x = \left[i_{sd}, E_{i_{sd}}, i_{sq}, E_{i_{sq}}, i_{gd}, E_{i_{gd}}, i_{gq}, E_{i_{gq}}, E_{P_g}, V_{dc}, E_{V_{dc}}, \beta, x_\beta, \omega_g, \omega_b, \omega_h, \theta_{sh-bh}, \theta_{sh-hg}, x_{edge}, \dot{x}_{edge}, x_{side}, \dot{x}_{side}, \theta_b\right]^T \\ z = \left[\omega_{g,ref}, V_{dc,ref}\right], u = \left[V_w, v_{gd}\right], 0 = g(x, z, d) \tag{26}$$

where $E_{i_{gd}}$, $E_{i_{gq}}$, E_{P_g}, $E_{i_{sd}}$, $E_{i_{sq}}$, $E_{V_{dc}}$ and E_β imply on the state variables of the integral components of the PI controllers related to the GSC and MSC currents, generated active power, DC-link voltage and pitch angle.

For modal analysis, the linearized representation of the whole studied WT system around an operating point can be expressed in the state-space representation as follows

$$\Delta\dot{x} = A.\Delta x + B.\Delta u \\ \Delta y = C.\Delta x + D.\Delta u \tag{27}$$

where A, B, C, and D are matrices with adequate dimensions related to the states, inputs, outputs, and the matrix that indicates the proportion of input directly reflecting on the output. During linearizing process, the operating points pertaining to the wind speed of 14 m/sec are selected as a test study. The reason for this is that the loading occurrence at the full-load area of WTs is more intense than the other WT areas [6,9,27], and [29]. Using (27) and the specifications of the NREL 5-MW reference WT model, as detailed in [30], the modal analysis of the studied WT system is carried out. A detailed list of the main parameters of the turbine-generator system under study is presented in the Appendix. Table 1 presents the system eigenvalues, corresponding frequencies, damping ratios, and dominant state variables associated with each mode.

Considering Table 1, it can be observed that the modes λ_{1516}, $\lambda_{17\text{-}18}$, $\lambda_{19,20}$, and $\lambda_{21,22}$ are known as the rotor in-plane modes corresponding to the dynamics of the drive-train, blade edgewise, and tower side-side, respectively, which intrinsically have poor damping ratios and will be dealt with in the next sections. The other modes generally related to the electrical modes, are well damped, and are out of scope of this study. Meanwhile, it is worth noting that the mechanical-structural modes, i.e. $\lambda_{15,16}$ and $\lambda_{19,20}$, have nearly similar natural frequencies, and thus the torsional oscillations can influence the edgewise deflections. This is because the tangential force, as the trigger factor for exciting the edgewise deflections, is explicitly impressed by the shaft twist angle $\theta_{sh\text{-}bh}$, which is relevant to the modes $\lambda_{15,16}$.

4. Modifying the basic control system for improving the rotor in-plane loading

To increase the WT operational lifetime, improving the rotor in-plane resonant loads should be taken into consideration. This aim can be realized by developing the WT control system, as addressed in this Section.

4.1. Selection of the most effective input–output signal pair for modifying the control system

For reducing the rotor in-plane loading and damping improvement of rotor in-plane oscillations, in this paper, the WT control system is modified by adding appropriate auxiliary control signals. In this way, in the first step, the suitable injecting points for applying the auxiliary control signals are identified by using controllability analysis, and then, by using the observability analysis, the appropriate feedback signals are chosen. It is noted that the modal controllability is related to the input signals and evaluates the effective modification sites to influence the desired modes via state feedbacks, also the modal observability refers to feedback signals and measures the visibility of the desired modes from the outputs. The controllability and observability matrices are given as [31,32]:

$$B^{'} = \Phi^{-1}B \\ C^{'} = C\Phi \tag{28}$$

where Φ is the right eigenvectors matrix (mode shape matrix), and B' and C' are the controllability and observability matrices, respectively.

According to Table 1, the state variables $\theta_{sh\text{-}bh}$ and $\theta_{sh\text{-}hg}$ have the most contribution in the modes, $\lambda_{15,16}$ and $\lambda_{17,18}$, respectively. In practice, the shaft twist angles, $\theta_{sh\text{-}bh}$ and $\theta_{sh\text{-}hg}$ are inaccessible for measuring and using them in the control system, nevertheless their dynamics can be

Table 2
Observability and Controllability Analyses for Different Feedback Signals and Modification Sites for Rotor In-plane Modes.

Modes	Feedback signals	Observability criterion	Modification sites	Controllability criterion
$\lambda_{15,16}$	$\omega_b-\omega_g$	0.0011	$P_{g,ref}$	690
			$i_{gd,ref}$	59
	$\omega_b-\omega_h$	0.001	$V_{dc,ref}$	35
			$i_{sq,ref}$	35
$\lambda_{17,18}$	$\omega_b-\omega_g$	0.0005	$P_{g,ref}$	240
			$i_{gd,ref}$	21
	$\omega_b-\omega_h$	0.0015	$V_{dc,ref}$	68
			$i_{sq,ref}$	68
$\lambda_{19,20}$	x_{edge}	0.0813	$P_{g,ref}$	350
			$i_{gd,ref}$	29
	$\dot{x}_{edge}$	0.9967	$V_{dc,ref}$	13
			$i_{sq,ref}$	13
$\lambda_{21,22}$	x_{side}	0.4442	$P_{g,ref}$	140
			$i_{gd,ref}$	43
	$\dot{x}_{side}$	0.8959	$V_{dc,ref}$	2
			$i_{sq,ref}$	2

expressed as a function of (ω_b-ω_g) or (ω_b-ω_h), as given in (29)-(32).

$$\left(D_{hg}+\frac{k_{hg}\omega_B}{s}\right)(\omega_b-\omega_g)-\left(k_{hg}+\frac{D_{hg}s}{\omega_B}\right)\theta_{sh-bh}-2H_g s\omega_g+T_e=0 \tag{29}$$

$$\left(D_{hg}+D_{hg}\frac{-2H_h s^2+D_{hg}s-k_{bh}\omega_B}{2H_h s^2+(D_{bh}-D_{hg})s+k_{bh}\omega_B}\right)(\omega_b-\omega_g)+\left(k_{hg}+\frac{k_{hg}D_{hg}s}{2H_h s^2+(D_{bh}-D_{hg})s+k_{bh}\omega_B}\right)\theta_{sh-hg}+\left(\frac{-2H_h D_{hg}s^2}{2H_h s^2+(D_{bh}-D_{hg})s+k_{bh}\omega_B}\text{ - }2H_g s\right)\omega_g+T_e=0 \tag{30}$$

$$\left(D_{bh}+\frac{2H_g D_{bh}}{k_{hg}\omega_B+D_{hg}s}\right)(\omega_b-\omega_h)+\left(k_{bh}+\frac{2H_g k_{bh}}{D_{hg}}\left(s-\frac{k_{hg}\omega_B s}{k_{hg}\omega_B+D_{hg}s}\right)\right)\theta_{sh-bh}-\left(2(H_h+H_g)s+\frac{2H_h H_g s^3}{k_{hg}\omega_B+D_{hg}s}\right)\omega_h+T_e=0 \tag{31}$$

$$\left(D_{bh}+\frac{\omega_B k_{bh}}{s}+\frac{2H_g\omega_B k_{bh}}{D_{hg}}+\frac{2H_g D_{bh}s}{D_{hg}}\right)(\omega_b-\omega_h)-2H_g k_{hg}\theta_{sh-hg}s-\left(2H_h s+\frac{2H_g s}{D_{hg}}(2H_h s+D_{hg})\right)\omega_h+T_e=0 \tag{32}$$

According to (29)-(32), the terms (ω_b-ω_g) and (ω_b-ω_h) can be chosen as feedback signals to add the damping torque and suppress the torsional oscillations. From Table 1, it can be observed that the edgewise and side-side vibrations are associated with their own deflections and velocities, x_{edge}, $\dot{x}_{edge}$, x_{side} and $\dot{x}_{side}$. These variables can be chosen as feedback signals by once or twice integrating the acceleration signals coming from accelerometers usually mounted on the blades and nacelle [3,12,13,20]. In summary, the suitable feedback signal to increase the damping of torsional modes $\lambda_{15,16}$ and $\lambda_{17,18}$ can be (ω_b-ω_g) or (ω_b-ω_h), and the appropriate feedback signals to increase the damping of the blade edgewise and tower side-side modes, i.e. $\lambda_{19,20}$, and $\lambda_{21,22}$, are x_{edge}, $\dot{x}_{edge}$, x_{side} and $\dot{x}_{side}$. The optimal feedback signals for adding the damping of the modes are obtained from the observability analysis. After choosing the appropriate feedback signals, the points corresponding to $P_{g,ref}$, $i_{gd,ref}$, $V_{dc,ref}$ and $i_{sq,ref}$ in Fig. 5 can be considered as candidates for applying the

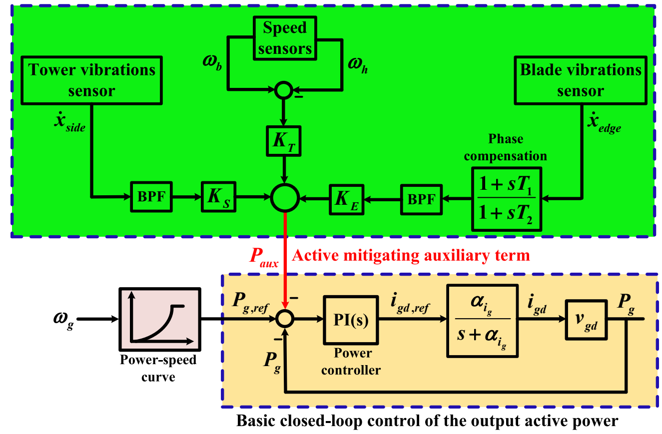

Fig. 7. Control structure of the proposed active mitigation approach.

feedback signals in the control system. The optimal point for applying the feedback signals is obtained from the controllability analysis. Table 2 presents the data achieved by the observability and controllability analyses for different feedback signals and modification points.

According to Table 2 and considering the entire rotor in-plane modes, the controllability indexes associated with the reference value of the WT generated active power, $P_{g,ref}$, have the highest values compared to the other candidates. Hence, this modification site is the best point to inject the auxiliary damping terms. Besides, the observability indexes strictly express that the best feedback signals for the active damping of the edgewise and side-side modes are the variables $\dot{x}_{edge}$ and $\dot{x}_{side}$, respectively. For the first torsional mode, $\lambda_{15,16}$, the observability index value corresponding to (ω_b-ω_g) is slightly higher than the one related to (ω_b-ω_h), whilst, for the second torsional mode, $\lambda_{17,18}$, (ω_b-ω_h) obviously has a higher observability index value than the one related to (ω_b-ω_g). Hence, considering the both torsional modes, $\lambda_{15,16}$ and $\lambda_{17,18}$, the feedback signal (ω_b-ω_h) can be chosen as the best feedback signal.

4.2. *Proposed control structure for active damping of the rotor in-plane modes*

Using the results obtained from Tables 1 and 2, the control structure of the GSC produced active power can be developed by using the optimal feedback signals and modifying the reference power $P_{g,ref}$, as depicted in

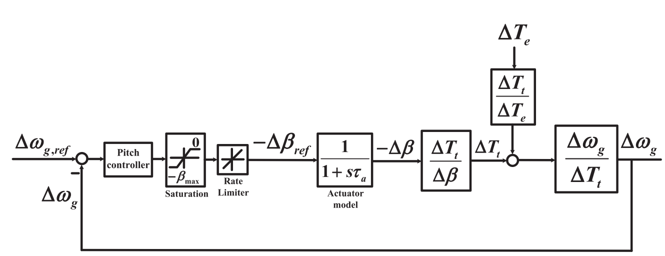

Fig. 8. Detailed control structure of the pitch system.

Table 3
Eigenvalues and Dominant State Variables for the WT with the Proposed Control Scheme.

Modes	Frequency (Hz)	Damping ratio	Dominant state variables
$\lambda_1 = -677.44$	–	1	i_{sd}
$\lambda_2 = -18.13$	–	1	$x_{i_{sd}}$
$\lambda_3 = -586.80$	–	1	i_{sq}
$\lambda_{4,5} = -628.32 \pm 0j$	–	1	i_{gd}, i_{gq}
$\lambda_{6,7} = -31.64 \pm 37.21j$	5.92	0.65	$V_{dc}, x_{V_{dc}}$
$\lambda_8 = -18.18$	–	1	$x_{i_{sq}}$
$\lambda_9 = -76.16$	–	1	x_{P_g}
$\lambda_{10,11} = -6.28 \pm 0j$	–	1	$x_{i_{gd}}, x_{i_{gq}}$
$\lambda_{12} = -9.05$	–	1	β
$\lambda_{13,14} = -0.78 \pm 0.43j$	0.068	0.88	$x_\beta, \omega_b, \theta_b$
$\lambda_{15,16} = -6.88 \pm 14.93$	2.38	0.42	θ_{sh-bh}, ω_g
$\lambda_{17,18} = -7.48 \pm 80.33j$	12.82	0.093	θ_{sh-hg}, ω_h
$\lambda_{19,20} = -0.18 \pm 12.08j$	1.92	0.015	$x_{edge}, \dot{x}_{edge}$
$\lambda_{21,22} = -0.16 \pm 2.04j$	0.32	0.078	$x_{side}, \dot{x}_{side}$

Fig. 7, where α_{ig} is the closed-loop bandwidth of the inner control loop of the GSC, and K_T, K_E, and K_S are the feedback gains. The band pass filters (BPFs) are placed in the feedback paths to extract the respective resonance components. According to Fig. 7, the auxiliary damping power, P_{aux}, comprises three components related to torsional, edgewise and side-side vibrations. This auxiliary damping term, indeed, adds a damping torque into the original electromagnetic torque via the generated active power control loop, resulting in the damping enhancement of the whole rotor in-plane modes.

In Fig. 7, the phase compensation unit, i.e. $1 + sT_1/1 + sT_2$, is used to produce an appropriate phase led in order to compensate for the phase-lag introduced in the trajectory from $\Delta P_{g,ref}$ to Δx_{edge}. To do this, we have

$$\frac{\Delta x_{edge}}{\Delta P_{g,ref}} = \frac{\alpha_P}{s+\alpha_P}\frac{\Delta f_{\tan}}{\Delta T_e}\frac{\Delta x_{edge}}{\Delta f_{\tan}} \tag{33}$$

where, α_P is the power closed-loop bandwidth. Because the transfer function of (33) pertains to the edgewise vibrations, then the frequency of interest for its design parameters is the edgewise natural frequency, 12.25 rad/sec, and in this frequency we have

$$\frac{\Delta x_{edge}}{\Delta P_{g,ref}}\Big|_{s=j12.25} = 80\angle -39.8^{\circ} \tag{34}$$

Based on (34), provided the edgewise velocity signal, $\dot{x}_{edge}$, by using the phase compensation unit, $1 + sT_1/1 + sT_2$, is advanced by 39.8° at the frequency of 12.25 rad/sec, the edgewise damping force will be in phase with $\Delta\dot{x}_{edge}$ and consequently will suppress the edgewise resonant loads.

A challenge that may arise from the proposed active mitigation approach is that the auxiliary term relevant to the side-side deflection can induce an additional burden with side-side harmonic contest on the torsional responses and, consequently, the tangential force. This drawback can be solved with redesigning the pitch control system. Fig. 8 depicts the detailed structure of the pitch control system, where the T_e behaves as the disturbance. Considering the proposed control approach of Fig. 7, and linearizing the generator torque, T_e, around an operating point, β_0, we have

$$\Delta T_e \cong \Delta P_g = G_1\Delta i_{sq} + G_2(\Delta\omega_b - \Delta\omega_h) + G_3\Delta\dot{x}_{edg} + G_4\Delta\dot{x}_{side} \tag{35}$$

In (35), the generator losses have been ignored, and G1 ~ G4 are defined as transfer functions linking the generator torque to the control variables. Thus, dynamics of the term $G_4\Delta\dot{x}_{side}$ in (35) can be appeared on the generator speed as a disturbance. In order to mitigate the side-side vibrations while attenuating the respective edgewise fluctuations, the following guidelines for designing the pitch control system should be considered:

(a) The higher wind speeds result in higher loop gains of $\Delta T_t/\Delta\beta$, and hence for the stability considerations, the pitch control system

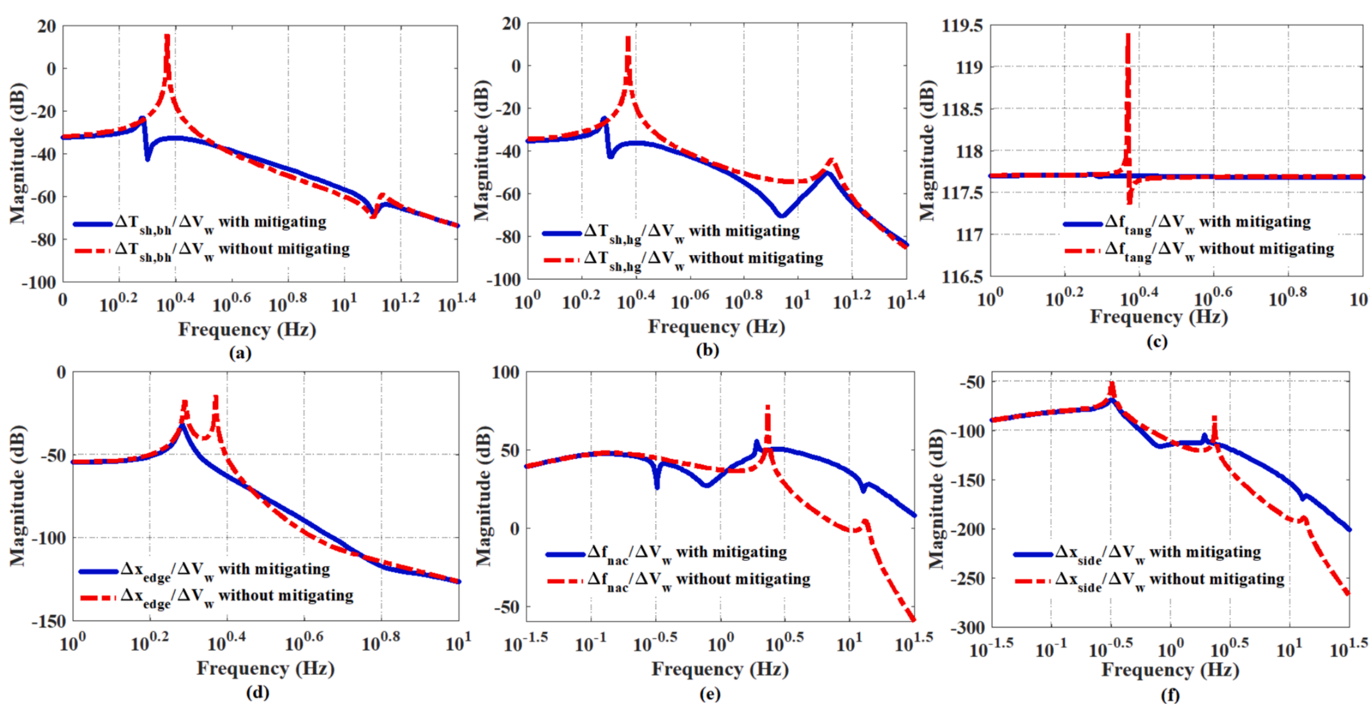

Fig. 9. The magnitude of the frequency responses of the torsional torques, tangential force, edgewise deflection, nacelle force, and side-side deflection to the wind speed for the situations with and without the proposed active mitigation method.

should be designed for an operating point with the worst case conditions.

(b) To ensure the stability constraints in the pitch control system, the open loop crossover frequency should be enough lower than the first order torsional frequency, in addition, to have an appropriate stability margin, the phase margin is set to 60 deg.

(c) To attenuate the effect of the side-side deflections on the generator speed response to the disturbance of generator torque, $\Delta\omega_g/\Delta T_e$, the pitch control closed-loop bandwidth should be chosen sufficiently lower than the tower side-side deflection frequency.

Table 3 draws the modal analysis results for the studied system with the proposed active mitigation method, wherein the operating points are associated with $V_w = 14$ m/sec. The feedback gains K_E, K_T and K_S are selected as 3/8, 10, and 15, respectively. Comparing Tables 1 and 3 it is comprehended that the proposed control approach can significantly improve the damping ratios of the rotor in-plane modes, such that the damping ratios of the torsional modes $\lambda_{15,16}$ and $\lambda_{17,18}$ are improved from 0.0183 and 0.0636 to 0.42 and 0.093, respectively. In addition, the damping ratios of the structural modes $\lambda_{19,20}$ and $\lambda_{21,22}$ are corrected from 0.005 to 0.015 and from 0.01 to 0.078, respectively. It is noted that the proposed control approach has not any unwanted effect on the electrical modes, especially the d-axis of the GSC current, modes $\lambda_{4,5}$ corresponding to the generated active power.

Fig. 9 discloses the magnitudes of the bode diagrams of the torsional torques, tangential force, edgewise deflection, nacelle force, and side-side deflection to the wind speed, V_w, for the situations with and without the proposed active suppression method. It is evident that under the resonant excitations caused by the aerodynamic perturbations, the diagrams associated with the proposed active mitigation method have smoother waveforms than the those without the active mitigating control. Meanwhile, the proposed active mitigation method prevents the impact of the torsional oscillations on the tangential force and the excitation of the edgewise modes.

5. Stability verification using Lyapunov-Based analysis

To theoretically validate the asymptotic stability of the studied WT system under the proposed active mitigating control approach, Lyapunov's indirect method is employed. This method provides a rigorous mathematical framework for verifying the stability of nonlinear systems by examining the local behavior of their linearized representations around an equilibrium point, without requiring explicit solutions of the system trajectories.

According to this theory, if the state matrix A of the linearized system is Hurwitz (i.e., all eigenvalues of A have strictly negative real parts), then for any given symmetric positive definite matrix Q, there exists a unique symmetric positive definite matrix P that satisfies the following Lyapunov equation:

$$A^TP + PA = -Q \tag{36}$$

A corresponding quadratic Lyapunov function can then be constructed as

$$V(x) = x^TPx \tag{37}$$

whose time derivative along the system trajectories is given by:

$$\frac{dV(x)}{dt} = -x^T(A^TP + PA)x = -x^TQx \tag{38}$$

Since Q is positive definite, this confirms that $V(x)$ is a valid Lyapunov function and the equilibrium point of the system is asymptotically stable. This result is consistent with classical stability theory, as thoroughly discussed in [33].

In this study, the state matrix A is derived from the small-signal model of the study WT. Eigenvalue analysis confirms that under both the basic and proposed control schemes, all eigenvalues of A exhibit negative real parts (Tables 1 and 3), satisfying the Hurwitz condition and thus proving asymptotic stability via Lyapunov's method.

However, although the basic control system guarantees closed-loop

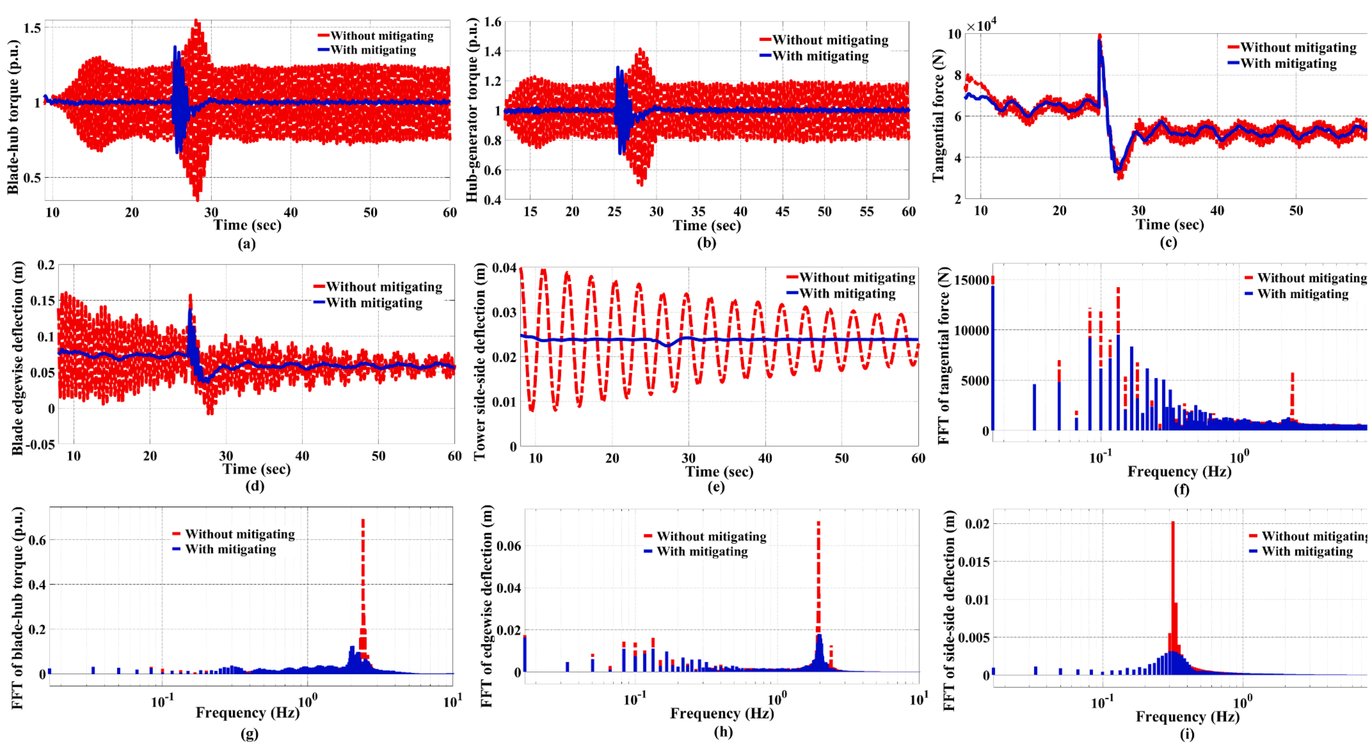

Fig. 10. Time domain simulations and frequency spectrums of the mechanical-structural responses under step turbulence of the wind speed for two cases with and without the proposed active mitigation method.

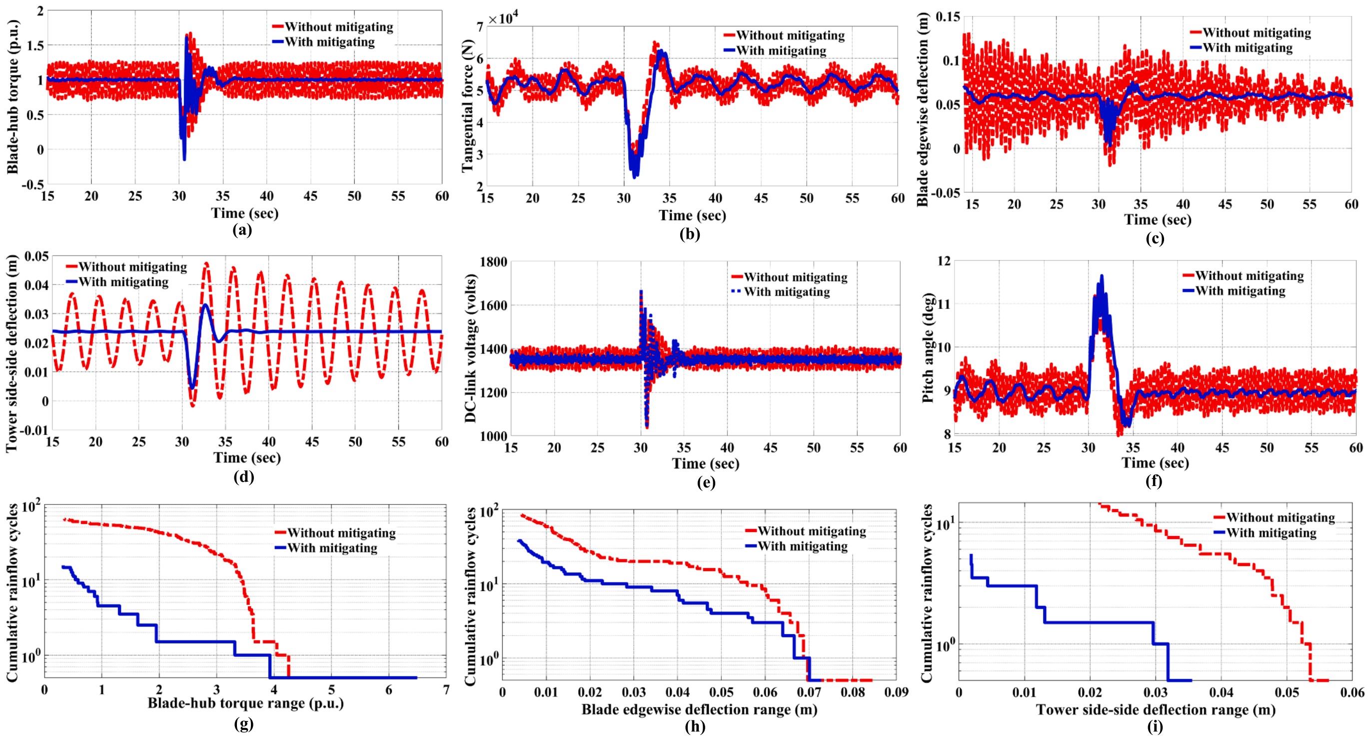

Fig. 11. Time domain simulations and cumulative rainflow cycles of the WT responses under grid fault occurrence for two cases with and without the proposed active mitigation method.

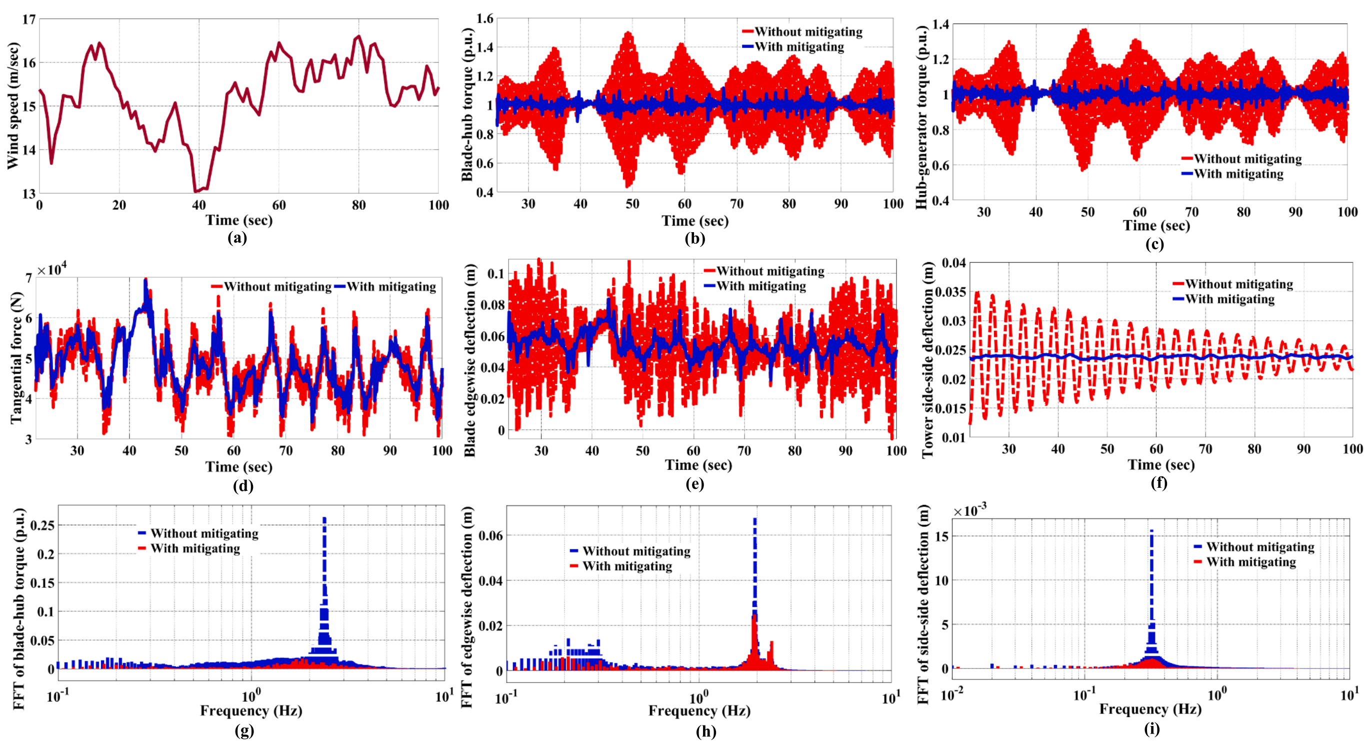

Fig. 12. Time domain simulations, frequency spectrums of the WT mechanical-structural responses under a wake-influenced wind field for two cases with and without the proposed active mitigation method.

stability, rotor in-plane modes, including torsional, blade edgewise, and tower side-side modes, exhibit low damping ratios, making them susceptible to oscillations under disturbances. To address this issue, the proposed active damping methos modifies the system dynamics such that the new state matrix retains the Hurwitz property while significantly improving the damping ratios of these critical modes (Table 3). This enhancement is reflected in the Lyapunov function, $V(x) = x^T P x$, where the revised P matrix results in faster decay rates (more negative $\dot{V}(x)$) for the previously weakly damped modes, without destabilizing other dynamics. Thus, the Lyapunov-based analysis not only confirms the asymptotic stability of the WT system with the proposed control scheme, but also highlights its effectiveness in enhancing the damping performance of the oscillatory modes in the WT system.

6. Simulation results

This section presents simulation results to evaluate the effectiveness of the proposed active damping control method using the nonlinear model of the studied wind turbine (WT) system. The simulation test bench is derived from an open-access MATLAB/Simscape wind turbine model, whose mechanical and structural characteristics have been adapted to match those of the NREL 5-MW reference WT. In addition, supplementary components from the SimWindFarm toolbox have been incorporated, as detailed in [34]. The resulting test bench, illustrated in Fig. 5, features a 5 MW, 690 V, 50 Hz PMSG-based WT connected to a 20 kV distribution grid via a 10 km, 20 kV cable and a step-up transformer, and includes all relevant control loops described in previous sections. The adopted modeling framework ensures adequate dynamic fidelity and modularity, supporting subsystem-level customization and access to internal variables required for control design and load analysis.

To assess dynamic performance, three case studies are examined: (i) wind speed step changes, (ii) a voltage dip at the generator terminals, and (iii) wind profile variations influenced by the wake effect. Three scenarios are simulated under two control schemes: the baseline case without mitigating, and the proposed control with active damping action.

Fig. 10 displays the time responses of the shaft torsional torques of the blade-hub and hub-generator shafts, tangential force, tip blade edgewise motions, and tower side-side deflections, respectively as (a)-(e), to an abrupt turbulence of the wind speed at t = 25 s from 12 m/sec to 14 m/sec. Meanwhile, for such conditions, the frequency spectrums of the tangential force, blade-hub shaft torsional torque, tip blade edgewise deflections, and tower side-side deflections are shown, respectively as (f)-(i), quantifying thevibrational responses by decomposing them into their frequency components.

It is observed that following the step increase in the wind speed from 12 m/s to 14 m/s at t = 25 s, both the tangential aerodynamic force and blade edgewise deflections decrease, despite the higher wind excitation. This seemingly counterintuitive behavior can be explained by the aerodynamic interaction between the inflow angle, pitch angle, and angle of attack. Specifically, the pitch actuation mechanism, which is activated to maintain constant power output in the full-load region, causes a rapid increase in blade pitch angle. As a result, the angle of attack decreases, leading to reduced lift and drag coefficients. Although the resultant wind velocity increases, the net aerodynamic load on the blades diminishes because the rate of increase in the pitch angle exceeds that of the inflow angle, thereby reducing the angle of attack. This ultimately lowers the lift and drag forces, and in turn, leads to a decrease in the tangential force and the associated blade edgewise deflections.

Fig. 11 exhibits the time domain simulations of the blade-hub torsional torque, tangential force, tip blade edgewise motions, tower side-side deflection, DC-link voltage, and pitch angle, respectively as (a)-(f), once an 80 % grid voltage sag is occurred at $t = 30$ s with duration of 500 msec. Besides, for such conditions, the cumulative rainflow cycles of the blade-hub torsion torque, tip blade edgewise deflections and tower side-side deflections are given, respectively as (g)-(i).

Fig. 12 (b)-(f) illustrates the time-series of the blade-hub shaft torsional torque, hub-generator shaft torsional torque, tangential force, blade flapwise motions, and tower side-side deflections under a wind field influenced by the wake effect, as shown in Fig. 12 (a). The corresponding frequency spectra are presented in (g)-(i). The wake effect is simulated for a wind farm comprising 10 WTs, configured in a network pattern with 150 × 170 m of spacing between adjacent turbines. A turbulence intensity of 0.1 and a mean wind speed of 15 m/s are applied in the model. This model is based on the revised IEC 61400–1 standard [28] and [35].

Considering Figs. 10-12, once the WT encounters to the wind speed turbulency either as step chang or wake variations, and even grid voltage fault, in the situation without the proposed active mitigation control, the intense transients are induced on the structural-mechanical waveforms of the WT, whereas in the situation with the active mitigation method, the loading intensity are considerably suppressed.

7. Conclusion

This paper proposes an active mitigation control method to resolve the rotor in-plane loads related to the blades, tower, and drive-trine in the large-scaled PMSG-based WTs. This approach has an efficient capability and a fairly simple mechanism to implement in practical applications. To design this approach, first of all, a comprehensive detailed model for precisely describing the dynamic features of the torsional, edgewise, and side-side vibrations is introduced. Based on this model and using small signal analysis, the oscillatory rotor in- plane modes and the respective dominant state variables are extracted. Secondly, different options for taking the feedback signals and injecting the auxiliary damping terms are investigated by the observability and controllability analyses, respectively. Thus, it is demonstrated that the signals related to the difference of the rotational speeds between the blade and hub, edgewise vibration velocity, and side-side vibration velocity are the effective feedback signals, besides, the power outer control loop is the best point to inject the auxiliary terms. Thirdly, after slightly correcting the feedback signal associated with edgewise vibrations and redesigning the pitch controller, the developed power control loop is obtained. Eventually, the simulation and analytical studies confirms that the proposed control method can efficiently alleviate the rotor in-plane loads under the wind and grid voltage disturbances.

CRediT authorship contribution statement

Ali Safaeinejad: Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Mohsen Rahimi:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. . Properties of WT model

Symbol	Quantity	Value
P_{rated}	Turbine rated power	5.23 MW
$Vw_{,n}$	Nominal wind speed	11.4 m/sec
$V_{w,cut\text{-}in}$	Cut-in wind speed	3 m/sec
$V_{w,cut\text{-}out}$	Cut-out wind speed	25 m/sec
ω_{rated}	Rated rotor speed	12.1 rpm

(*continued on next page*)

(*continued*)

Symbol	Quantity	Value
R	Blade radius	63 m
n_g	Gearbox ratio	62
ρ	Air density	1.222 kg/m^3
J_b	Blade moment of inertia	2.84×10^7 kgm^2
J_h	Hub moment of inertia	753,519 kgm^2
H_{tow}	Hub height	90 m
M_{tow}	Effective mass of the tower	6.975×10^5 kg
K_{tow}	Stiffness coefficient of the tower	2.84×10^6N/m
D_{tow}	Tower damping ratio	1 %
$K_{sh\text{-}bh}$	Blade-hub shaft stiffness coefficient	6.6×10^8N.m/rad
$K_{sh\text{-}hg}$	Hub-generator shaft stiffness coefficient	3.66×10^9N.m/rad
a	Tower radius	2.4 m
α	Wind shear component	0.3
x	Blade origin from tower midline	5 m
τ_β	Time constant of the pitch actuator	0.1 sec
M_{ege}	Mass of each blade	5913.3 Kg
K_{edge}	Stiffness coefficient of the blades	2.8247×10^5N/m
D_{edge}	Edgewise damping ratio	0.48 %

Appendix B. . Properties of PMSG

Symbol	Quantity	Value
S_b	Generator rated power	5 MVA
V_b	Rated voltage	690 V
n_p	Pole pairs	4
ω_B	Base angular electrical frequency	$100 \times \pi$ rad/sec
J_g	Generator moment of inertia	2.12×10^6 kgm^2
R_s	Stator resistance	8.28×10^{-4} Ω
L_s	Stator inductance	4.091×10^{-5}H
ψ_{pm}	Permanent magnet flux linkage	1.7036 Wb

Data availability

No data was used for the research described in the article.

References

[1] Fitzgerald B, Basu B. Vibration control of wind turbines: Recent advances and emerging trends. Int. J. Sustain. Mater. Struct. Syst Nov. 2020;4:347–72.
[2] H. Zuo, K. Bi and H. Hao, “A state-of-the-art review on the vibration mitigation of wind turbines”, *Renewable Sustain. Energy Rev.*, vol. 121, Apr. 2020.
[3] Zhang Z, Nielsen SRK, Blaabjerg F, Zhou D. Dynamics and control of lateral tower vibrations in offshore wind turbines by means of active generator torque. Energies 2014;7(11):7746–72.
[4] Mozafari M, Fathi M, Jafarian M. An integrated CFD-FEM approach for analyzing fluid–structure interaction in offshore wind turbines. Thin-Walled Struct 2024;194:113113.
[5] Guma G, et al. Aero-elastic analysis of wind turbines under turbulent inflow conditions. Wind Energy Sci. Discuss. 2020;2020:1–20.
[6] Ismaiel A. A multivariate machine learning approach for the prediction of wind turbine blade structural dynamics. Appl. Syst. Innov. 2025;8(1):12.
[7] J. Ao, Y. Li, S. Hu, S. Gao, and Q. Yao, “Data-driven modeling for wind turbine blade loads based on deep neural network,” *Energy Eng.*, vol. 121, no. 12, 2024.
[8] F. Xie, A. Aly, “Structural control and vibration issues in wind turbines: A review,” *Engineering Structures*, vol. 210, pp. 0141-0296, May. 2020.
[9] Rahimi M. Improvement of energy conversion efficiency and damping of wind turbine response in grid connected DFIG based wind turbines. Int. J. Elec. Power 2018;95:11–25.
[10] Rahimi M, Beiki A. Efficient modification of the control system in PMSG-based wind turbine for improvement of the wind turbine dynamic response and suppression of torsional oscillations. Int Trans Electr Energy Syst Aug. 2018;28(8):2578.
[11] Beiki A, Rahimi M. An efficient sensorless approach for energy conversion enhancement and damping response improvement in permanent magnet synchronous generator (PMSG) based wind turbines. Int Trans Electr Energy Syst Jul. 2019;29(1):2684.
[12] Y. Wang, Y. Guo, D. Zhang, H. Liu and O. Song, “Analysis and mitigation of the drive train fatigue load for wind turbine with inertial control, ”*International Journal of Electrical Power & Energy Systems*, vol. 136, pp. 107698, March. 2022.
[13] Kavil Kambrath J, Khan MSU, Wang Y, Maswood AI, Yoon Y-J. A Novel Control Technique to Reduce the Effects of Torsional Interaction in Wind Turbine System. IEEE Journal of Emerging and Selected Topics in Power Electronics, Sept. 2019;7(3):2090–105.
[14] Z. Li, S. Tian, Y. Zhang, H. Li, and M. Lu, “Active Control of Drive Chain Torsional Vibration for DFIG-Based Wind Turbine,” *Energies*, vol. 12, no. 9, pp. 1744, May. 2019.
[15] Prajapat GP, Senroy N, Kar IN. Wind Turbine Structural Modeling Consideration for Dynamic Studies of DFIG Based System. IEEE Trans Sustainable Energy Oct. 2017;8(4):1463–72.
[16] Staino A, Basu B. Dynamics and control of vibrations in wind turbines with variable rotor speed. J. Eng. Struct. Nov. 2013;56:58–67.
[17] Staino A, Basu B, Nielsen SRK. Actuator control of edgewise vibrations in wind turbine blades. J Sound Vib Mar. 2012;331:1233–56.
[18] Licari J, Ekanayake JB, Jenkins N. Investigation of a Speed Exclusion Zone to Prevent Tower Resonance in Variable-Speed Wind Turbines. IEEE Trans Sustainable Energy Oct. 2013;4(4):977–84.
[19] Mohammadi E, Fadaeinedjad R, Moschopoulos G. Implementation of internal model based control and individual pitch control to reduce fatigue loads and tower vibrations in wind turbines. J Sound Vib May 2018;421:132–52.
[20] Abo-Khalil AG, Alyami S, Sayed K, Alhejji A. Dynamic modeling of wind turbines based on estimated wind speed under turbulent conditions. Energies May 2019;12(10):1907.
[21] Rahimi M. Mathematical modeling dynamic response analysis and control of pmsg-based wind turbines operating with an alternative control structure in power control mode. Int Trans Electr Energy Syst 2017;27(12):e2423.
[22] Van der Hooft E, Schaak P, Van Engelen T. Wind turbine control algorithms 2003.
[23] Burton T, Sharpe D, Jenkins N, Bossanyi E. Wind Energy Handbook. Chichester, K: Wiley; 2011.
[24] Jia F, Cai X, Lou Y, Li Z. Interfacing technique and hardware-in-loop simulation of real-time co-simulation platform for wind energy conversion system. IET Gener. Transmiss. Distrib. Sep. 2017;11(12):3030–8.
[25] Safaeinejad A, Rahimi M, Zhou D, Blaabjerg F. A sensorless active control approach to mitigate fatigue loads arising from the torsional and blade edgewise vibrations

in PMSG-based wind turbine system. Int J Electr Power Energy Syst Oct. 2023;155: 109525.

[26] Safaeinejad A, Rahimi M, Zhou D, Blaabjerg F. An efficient power control strategy for active mitigation of blade in-plane fatigue loading in PMSG-based wind turbines. IET Renew Power Gener Jan. 2024;18(2):261–82.

[27] Liu H, Tang Q, Chi Y, Zhang Z, Yuan X. Vibration reduction strategy for wind turbine based on individual pitch control and torque damping control. Int. Trans. Electr. Energ. Syst. Oct. 2016;26:2230–43.

[28] A. Safaeinejad, M. Rahimi, D. Zhou, and F. Blaabjerg, "Pitch control scheme considering entire dynamics and full-load region in PMSG-based wind turbines," *IEEE Transactions on Sustainable Energy*, pp. 1–13, Jan. 2024.

[29] Licari J, Ugalde-Loo CE, Liang J, Ekanayake J, Jenkins N. Torsional damping considering both shaft and blade flexibilities. Wind Eng 2012;36:181–96.

[30] J. Jonkman, S. Butterfield, W. Musial and G. Scott, *Definition of a 5-mw reference wind turbine for offshore system development*, Feb. 2009.

[31] Pal B, Chaudhuri B. Robust Control in Power Systems, New York, NY. USA: Springer; 2006.

[32] P. Kundur, Power System Stability and Control, McGraw-hill, vol. 7, 1994.

[33] Khalil HK. *Nonlinear Systems*. 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall; 2002.

[34] Grunnet JD, et al. AEOLUS toolbox for dynamic wind farm model simulation and control. Proc. Eur. Wind Energy Conf. Exihib. 2010:1–10.

[35] Wang L, Dong M, Yang J, Wang L, Chen S, Duić N, et al. Wind turbine wakes modeling and applications: Past, present, and future. Ocean Eng Oct. 2024;309(1): 118508.